Dr Lucy **Ham**

CV

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Education

December 2017 – Oct 2013 | Doctor of Philosophy (Pure Mathematics) | La Trobe University , Melbourne, Australia

Thesis: Definability and Complexity: Universal Horn classes of finite structures

- Supervisors: Prof Brian Davey and Prof Tomasz Kowalski
- Established a novel promise-problem approach to classifying the complexity of constraint satisfaction problems
- Passed with no amendments, five publications including three sole-authored.

July 2013 – Jan 2012 | Bachelor of Science (Honours in Mathematics) | La Trobe University, Melbourne, Australia

Thesis: Residual Bounds of Algebras and Boolean Topological Algebras

- Highest overall performance in the Faculty of Science
- First Class Honours

Oct 2011 – March 2008 | Bachelor of Science | La Trobe University, Melbourne, Australia

- Major in Mathematics and Statistics
- Weighted Average Mark (WAM): 95%

Experience

Planned 2027 | Visiting Research Scholar | Stony Brook University, Stony Brook, NY, USA

Extended research visit hosted by Prof Ramon Grima

- Planned extended research visit developing stochastic and mechanistic modelling approaches for cellular systems

Present – March 2024 | Senior Research Fellow, MACSYS | The University of Melbourne, Melbourne, Australia

ARC Centre of Excellence for the Mathematical Analysis of Cellular Systems (led by Prof Jennifer Flegg)

- Lead the development of a cross-node project on optimisation for biological design, with potential industry translation partner Samsara Eco
- Contribute to the training and professional development of students and ECRs
- Equity, People & Culture committee member, promoting inclusive culture and fair processes across the centre

March 2024 – Nov 2018 | Postdoctoral Research Fellow | The University of Melbourne, Melbourne, Australia

Theoretical Systems Biology Group (led by Prof Michael Stumpf)

- Change of field from Pure Mathematics to Mathematical Systems Biology
- Nine publications (eight first-authored) in high-impact journals such as *Physical Review Letters*, *Nature Communications*, *Cell Systems*, *Journal of Chemical Physics*, and *eLife*

Nov 2018 – Sept 2013 | Casual Demonstrator and Support Tutor | La Trobe University, Melbourne, Australia

School of Engineering and Mathematical Sciences

- Instructed and oversaw classes of up to 30 students for a range of first-year and second-year undergraduate university subjects including Discrete Mathematics, Calculus and Differential Equations, Applied Algebra and Number Systems
- Improve overall engagement, retention, success and well-being of first-year undergraduate students

Awards and Academic Distinctions

2026 **Most Persuasive Pitch** (MACSYS Annual Retreat Research Pitch Competition; three-minute research translation pitch, open to all Centre students and Research Fellows, ARC Centre of Excellence for the Mathematical Analysis of Cellular Systems)
2017 **Architect Michael Kaufman Scholarship*** (awarded to the highest achieving student entering the Master of Architecture at the University of Melbourne) (*Declined)
2013 **D.M. Myers University Medal** (awarded to the most outstanding undergraduate student graduating in the Faculty of Science across all campuses)
2013 **Prof C.J. Eliezer Prize in 4th Year Mathematics** (highest overall mark in fourth year mathematics)
2012 **Third Year Prize in Statistics** (highest overall mark in third year statistics)
2012 **E.J. Hannan Prize for Third Year of Accredited Major in Statistics** (highest overall mark in third year statistics for students taking a major in both statistics and mathematics)
2012 **La Trobe Alumni Prize** (best overall academic performance in the final year of their degree university-wide)
2011 AMSI Vacation Research Scholarship
2011 **National Bank of Australia Prize in 3rd Year Mathematics** (highest overall mark in third year mathematics)
2010 **Second Year Prize in Statistics** (highest overall mark in second year statistics)
2008 – 2010 **Dean's Honours List**

Publications

My research sits at the interface of applied mathematics and complex biological systems. I develop analytical and computational frameworks for understanding how stochasticity shapes cellular behaviour, from noise propagation in gene regulatory networks to emergent decision making in multicellular systems. Of my 17 publications to date, 15 are first-authored and 1 is senior-authored, with work appearing in premier journals including *Physical Review Letters*, *Nature Communications* (×2), *eLife*, *Cell Systems*, and *Journal of Chemical Physics* (all top 5% in at least one ScimagoJR category).

Under review

2026 **L. Ham**, M. Jackson, A. Sukys, M. P. H. Stumpf. *Intercellular signaling drives robust cell fate and patterning in multicellular systems*, bioRxiv:10.1101/2025.10.09.681513.

To appear

2026 A. Sukys*, **L. Ham***, K. Öcal, J. Holehouse, M. Ma, R. Grima, *Modeling gene expression noise: from transcriptional dynamics to cell growth and division*, *Current Opinion in Systems Biology*.* Joint first authorship.
2026 **L. Ham**†, M. Jackson†. *Finite model theory for pseudovarieties and universal algebra: preservation, definability and complexity*, arXiv:2212.02653 (*The Journal of Symbolic Logic*).† Equal authorship

Refereed journal publications

2026 Y. Liu, A. Zanca, M. P. H. Stumpf, **L. Ham**. *Molecular noise modulates transitions in the cell-fate differentiation landscape*, *npj Systems Biology and Applications*, doi:10.1038/s41540-026-00762-8.
2025 C. D. Madsen, A. Barbensi, S. Y. Zhang, **L. Ham**, A. David, D. E. V. Pires, M. P. H. Stumpf. *The topological properties of the protein universe*. *Nature Communications* **16**(1):7503.
2025 **L. Ham***, T. E. Woodward*, M. A. Coomer, M. P. H. Stumpf. *Mapping, modeling, and reprogramming cell-fate decision-making systems*. *Annual Review of Biomedical Data Science* **8**:537–562. * Joint first authorship.
2024 **L. Ham**, M. A. Coomer, K. Öcal, R. Grima, M. P. H. Stumpf. *A stochastic vs deterministic perspective on the timing of cellular events*. *Nature Communications*. bioRxiv:2023.07.20.549956.
2022 **L. Ham***, M. A. Coomer*, M. P. H. Stumpf. *The chemical Langevin equation for biochemical systems in dynamic environments*. *Journal of Chemical Physics* **157**(9):094105. *Joint first authorship.
2022 M. A. Coomer*, **L. Ham***, M. P. H. Stumpf. *Noise distorts the epigenetic landscape and shapes cell-fate decisions*. *Cell Systems* **13**(1):83–102. Selected for issue cover (original illustration) and featured on UniMelb Newsroom. *Joint first authorship.
2021 **L. Ham**, M. Jackson, M. P. H. Stumpf. *Pathway dynamics can delineate the sources of transcriptional noise in gene expression*. *eLife* **10**:e69324.

2020 **L. Ham**, D. Schnoerr, R. D. Brackston, M. P. H. Stumpf. *Exactly solvable models of gene transcription*. *Journal of Chemical Physics* **152**:144106. Cover article; selected for "Highlights 2020" and featured in *Scilight*.

2020 **L. Ham***, R. D. Brackston*, M. P. H. Stumpf. *Extrinsic noise and heavy-tailed laws in gene expression*. *Physical Review Letters* **124**:108101. *Joint first authorship.

2018 **L. Ham**, M. Jackson. *Axiomatisability, preservation and hardness for hypergraph quasivarieties*. *Algebra Universalis* **79**(2):1–17.

2017 **L. Ham**. *Gap theorems for robust satisfiability: Boolean CSPs and beyond*. *Theoretical Computer Science* **676**:69–91.

2017 **L. Ham**. *Relativised homomorphism preservation at the finite level*. *Studia Logica* **105**:761–786.

Refereed conference proceedings

2017 **L. Ham**†, M. Jackson†. *All or Nothing: toward a promise-problem dichotomy for constraint problems*. *Principles and Practice of Constraint Programming (CP 2017)*, LNCS **10416**:139–156. (CORE A). † Equal authorship.

2016 **L. Ham**. *A gap trichotomy for Boolean constraint problems: extending Schaefer's Theorem*. *ISAAC 2016*:36:1–36:12. (CORE A).

Selected manuscripts in preparation

In progress

- *Clonal heterogeneity and treatment response in cancer immunotherapy*. Joint work with J. Kuzich, Y. S. Foo, A. Sukys, A. Zanca and M. Dawson.
- *Quantifying spatial variability in gene expression from spatial transcriptomic data*. Joint work with K. Öcal, A. Sukys and R. Grima.
- *Cell-fate biases in human gastruloids*. Joint work with P. Konsa, Y. Liu, L. Diaz, N. Moris and M. P. H. Stumpf.
- *A hydrodynamics view of Waddington landscape* Joint work with Y. Liu, A. Zanca and M. P. H. Stumpf.
- *The gradient constraint: the incompatibility between non-equilibrium chemical reaction networks and epigenetic landscape models of development*. Joint work with A. Sukys and M. P. H. Stumpf.
- *Stochastic versus deterministic wave-front propagation*. Joint work with A. Zanca, A. Sukys and M. Jackson.

Teaching Experience

Course coordination and lecturing

SM1 2025 **Mathematics and Statistics for Biomedicine (MAST10024, undergraduate level 1), University of Melbourne**

- Coordinator and lecturer of a large first-year service subject enrolling over 80 students.
- Led a teaching team of tutors and demonstrators; oversaw delivery of lectures, practicals, and assessments.
- Incorporated research-led content, embedding applied mathematics into biomedical contexts (e.g. whole cell modelling for bio-production in industry).
- Taught statistical and computational tools (hypothesis testing, regression, epidemic modelling) to demonstrate real-world applications of mathematics in biomedicine.
- Student evaluations (n=34) rated the subject highly for intellectual engagement (4.06/5.00) and clear expectations (4.21/5.00).
- Student comments included "Lucy was a super helpful presence in this course", "very good with explaining clearly and going at a good pace", "very approachable", and from mid-semester feedback, "Lucy is inspiring and clearly passionate about the subject".

2023 **Thinking Scientifically (SCIE20001, undergraduate level 2), University of Melbourne**

- Supported delivery of the data science module in one of the university's largest second-year breadth subjects (800+ students).
- Redesigned learning exercises and statistical labs, modernising Minitab worksheets based on real-world datasets.
- Provided subject support and student engagement through Ed Discussion (an online forum for the subject), addressing queries and clarifying key concepts.

Workshop leadership

2026 **Fundamentals of Biochemical Modelling, MIG Workshop, 7 October**
- Workshop providing foundational training in biochemical modelling.

Mechanistic Inference from Single-Cell Data, MIG Workshop, 4 November
- Workshop on mechanistic approaches to inference from single-cell biological data.

2025 **EMBL-Australia PhD Training course**
- Provided foundational training across modern quantitative and experimental approaches in biology for early-stage PhD students.

2024 **The Fundamentals of Whole-Cell Modelling: Stochastic Simulation in Julia, BioInfoSummer Workshop, University of Melbourne**
- Developed and delivered interactive, hands-on workshop materials for a national audience of students and ECRs.
- Workshop GitHub Repository (link)
- Introduced Julia programming for stochastic simulation of biochemical systems, linking state-of-the-art research with practical computational skills.
- Received strong positive feedback for clarity, accessibility, and relevance of materials.

Tutoring experience

2013–2018 **Tutor and Maths Hub Subject Support, La Trobe University**
- Delivered tutorials, problem-solving sessions, and individual support across a wide range of undergraduate mathematics and statistics subjects, from first-year service teaching to advanced honours topics.
- **Mathematics:** Discrete Mathematics (L1), Calculus and Differential Equations (L1), Number Systems (L1), Mathematics for Biology (L1), Applied Algebra (L2), Vector Calculus (L3), Complex Analysis (L3), Model Theory (L4).
- **Statistics:** Applied Statistics (L2), Probability Models (L2), Biostatistics (L3).
- Consistently received positive student feedback for clarity of explanations, enthusiasm, and supportive engagement
- L1–L4 denotes undergraduate levels 1 to 4 respectively

Supervision and Mentoring

PhD supervision

2019–2022 **Dr Megan Coomer**, University of Melbourne (Co-supervisor with Prof Michael Stumpf)
- **Thesis:** Unravelling complex molecular networks that drive stochastic cell-fates
- **Current position:** Co-founder and Chief Executive Officer, Cell Bauhaus (Gates Foundation-backed biotechnology venture)
- Lead author publication in *Cell Systems* (2022)
- Presented at multiple national conferences
- I provided day-to-day and technical supervision, encompassing research design, model development, data analysis, and manuscript preparation

2023–present **Yujing Liu**, University of Melbourne (Co-supervisor with Prof Michael Stumpf)
- Senior-authored publication with Yujing
- Successfully passed third-year review milestone

2023–present **Taylor Woodward**, University of Melbourne (Co-supervisor with Prof Michael Stumpf)
- Joint first-author publication in *Annual Review of Biomedical Data Science* (2025)
- Successfully passed confirmation milestone

Research assistant supervision

2024-2025 **Tom Cummings, Yang Zhao and Joel Maldonado**
- Co-mentored research assistants on the project titled *Clone wars: predicting clonal diversity in resistance to CAR-T cell immunotherapy*
- Students jointly presented the outcomes of this project at the ANZIAM conference in February 2025

Research grants and funding

Current and recent grants

2026 **MACSYS Strategic Research Grant Scheme**
- Project Lead, MechBO: Mechanistically informed Bayesian Optimisation, a framework embedding mechanistic priors (kinetic, thermodynamic, structural) into Gaussian process surrogates for biological design (**$166,793**)
- Cross-institutional collaboration between UoM, ANU and Monash University
- Experimental validation partnership with the Australian Genome Foundry (Macquarie University)
- In active discussion with Samsara Eco, an Australian biotech company developing enzymatic plastic recycling at industrial scale, for industry partnership

2025 **Small Grants Scheme**, School of Mathematics and Statistics, The University of Melbourne **($5,000)**
2024 **Seed Funding Grant**, Melbourne Mathematical Biology, The University of Melbourne (**$20,000)**
2024 **BioSciences Early Career Researcher Grant**, The University of Melbourne (**$2,000)**
2024 **Strategic Initiative Travel Grant**, MACSYS (**$12,000)**

PhD and early career grants

2018 **AMSI Choose Maths Travel Grant** - Full support to participate in BioInfoSummer, University of Western Australia ($1,000)
2017 **Prof Ed Smith Travelling Scholarship** - Awarded to a full-time PhD student of outstanding merit to spend significant time overseas to pursue research ($10,000)

Grant proposals under development

2026 **NHMRC Synergy Grant (in development)**:
- Co-investigator with Melbourne Integrative Genomics, Peter MacCallum Cancer Centre, Prof Michael Stumpf
- Developing integrative frameworks for understanding tumour evolution through integration of cancer biology, mathematical modeling, and statistical bioinformatics
- Narrowly missed funding, scoring within 0.002 of the funding cutoff
- Highly ranked; team currently refining and resubmitting for the next round

2026 **ARC Future Fellowship Scheme (NOI submitted)**:
- The mathematical foundations of engineering living systems: develop stochastic and data-driven models for scalable and robust pipelines for synthetic biology and biotechnology.
- Partnership with Australian Genome Foundry on rational design of engineered cellular systems

2026 **Human Frontier Science Program (planning)**: New Investigator proposal with Dr Chris Miles (University of Utah), Dr Naomi Moris (The Francis Crick Institute), and Dr Christoph Zechner (SISSA Trieste)

Research impact and recognition

Invited presentations

2026 **Invited mini-symposium speaker** at the Annual Meeting of the Society for Mathematical Biology, Whole Cell Modelling: perspectives and progress, Graz, Austria
2025 **Invited seminar speaker** at the EMBL-Australia PhD course, SAiGENCI, Adelaide, Australia
2025 **Invited mini-symposium speaker** at the Annual Meeting for the Society of Mathematical Biology, The unexpected consequences of stochasticity in cell biology, Edmonton, Canada
2024 **Invited speaker** at Modeling and Inference of Stochastic Processes in Cells, Flatiron Institute, New York.
2024 **Keynote invited speaker** at CSIRO's Annual Conference for Advanced Engineering FSP, Fortitude Valley, Brisbane, Australia.
2023 **Keynote invited speaker** at Annual Conference for the Australian Research Council Centre of Excellence in Synthetic Biology, Sunshine Coast, QLD, Australia.
2022 **Invited speaker** at 12th European Conference on Mathematical and Theoretical Biology, Mini-Symposium on Stochastic Modelling of Genetic and Biochemical Systems, Heidelberg, Germany
2022 **Invited plenary speaker** at VicANZIAM, Melbourne, Australia
2022 **Invited plenary speaker** at BioInfoSummer 2022, The University of Melbourne, Australia
2022 **Invited speaker** at the British Applied Mathematics Conference, Loughborough University, UK

Conference and seminar presentations (selected)

2024 Seminar speaker, Elowitz Lab, Caltech, LA
2024 Seminar speaker, Covert Lab, Stanford, CA

2023 Speaker at the annual conference of the Australian Mathematical Society, The University of Queensland, Brisbane, Australia
2022 Speaker at the Biomathematics seminar, Imperial College, London, UK
2022 Speaker at the Systems and Synthetic Biology seminar, University College London, London, UK
2021 Shaping the Epigenetic Landscape: Complexities and Consequences, Society for Mathematical Biology Annual Meeting, Virtual on behalf of University of California Riverside (UCR)
2020 Sounding out the noise, Theoretical Systems Biology research seminar, The University of Melbourne
2019 Extrinsic noise in gene transcription: models and heavy-tailed laws, 63rd Annual Meeting of the Australian Mathematical Society, Monash University, Melbourne
2016 A gap trichotomy for Boolean constraint problems: extending Schaefer's Theorem, ISAAC 2016: The 27th Symposium on Algorithms and Computation, Sydney
2016 A gap trichotomy for Boolean constraint problems, 34th Victorian Algebra Conference, La Trobe University, Melbourne
2016 A gap trichotomy for Boolean constraint problems, 17th International Workshop on Logic and Computational Complexity, Aix-Marseille Université, France
2016 Mind the gap, 60th Annual Meeting of the Australian Mathematical Society, Australian National University, Canberra
2015 Definability of SP-classes of uniform hypergraphs, 59th Annual Meeting of the Australian Mathematical Society, Flinders University, Adelaide
2014 Preservation theorems on finite structures: algebra vs relations, 32nd Victorian Algebra Conference, Monash University, Melbourne
2014 Preservation theorems on finite structures: algebra vs relations, 38th ACCMCC, Victoria University of Wellington, New Zealand

Leadership and Service

Editorial and review activities

2019–present **Reviewer** for *Physical Review Letters*, *Journal of Chemical Physics*, *Biophysical Journal*, *European Journal of Applied Mathematics*, *Current Opinion in Systems Biology*, and others

Committee service and professional leadership

2026–present **Equity, Diversity and Inclusion (EDI) Committee Member, Australian Mathematical Society (AustMS)** Contributing to national initiatives promoting equity, diversity, inclusion and inclusive practices across the Australian mathematical sciences community.
2019, 2024, 2025 **B. H. Neumann Prize Committee Member** for the Australian Mathematical Society – Judge for prestigious student prize at annual national conference
2025-present **Early Career Researcher Representative** – Equity, People and Culture Committee (EPCC), ARC Centre of Excellence MACSYS
2025 **Interview Selection Panels for MACSYS and Cell Bauhaus**
2024 **Early Career Researcher Representative** – Operations Committee (OC), ARC Centre of Excellence MACSYS

Conference organisation

2027 **Organising Committee Member, MATRIX Research Program** – *Modelling and Inference for Cellular Systems*, MATRIX, Creswick, Victoria
- Contributing to the scientific program and organisation of a five-day research workshop bringing together researchers in mathematical biology, stochastic modelling, statistical inference and machine learning.

2026 **SMB Mini-symposium Organiser, Society for Mathematical Biology Conference** – Mechanistic Inference of Stochastic Single-Cell Dynamics
- Attracted keynote speaker Prof Ramon Grima (Stony Brook, NY) and seven other speakers including from Caltech, Edinburgh and Imperial College London

2024 **Theme Organiser, AMSI BioInfoSummer Conference** – Whole Cell Modelling theme
- Attracted keynote speakers Prof Markus Covert (Stanford) and Dr Megan Coomer (Cell Bauhaus)
- Co-coordinated and delivered a workshop on stochastic modelling in Julia

2023 **Co-organiser, Theory of Living Systems and MACSYS Webinar Series**
- Curated talks by leading global researchers in whole-cell modelling and in silico biology

2022 **Judge** for student prizes at Australian Biophysics Society meeting and Australian Algebra Conference

2016/17 **Assistant Organiser** of workshops and conferences:
- Universal Structures in Mathematics and Computing Workshop, La Trobe University
- 34th Victorian Algebra Conference, La Trobe University
- General Algebra and its Applications Conference, La Trobe University

2017 **Marker** for the Simon Marais Mathematics Competition

Public engagement and outreach

Arts and science integration

2026 **MACSYS Design Competition Winner**
- Designed a logo translating the Centre's core science into a visual identity
- Design to be printed on MACSYS merchandise and used across Centre communications and events

2025–2026 **Assistant coordinator and collaborator, MACSYS Artist-in-Residence Program**
- Assisted with the coordination of a centre-wide residency developed in partnership with the Monash University Museum of Art (MUMA), connecting contemporary artists with researchers to create public-facing work inspired by MACSYS research.
- Served on the artist-selection panel, reviewing applications and interviewing shortlisted candidates.
- Collaborating with Artist-in-Residence Mia Salsjö on a proposal for a joint exhibition at an experimental contemporary-art space, translating ideas from mathematics into visual and spatial works.

Media and science communication

2026 **Shortlisted for Superstars of STEM program** Science & Technology Australia's national program building the public profile and media skills of women and non-binary experts in STEM

2026 **Outreach panellist**, MACSYS Theme 4 workshop: "Translating whole-cell and virtual-cell models into real biological impact", Bio21 Institute, The University of Melbourne, AU

2024 **Featured in CSIRO's Cosmos magazine article**: *Subtracting myths out of maths* – on demystifying mathematics and making it accessible to diverse audiences (link)

2023 **Co-author, popular-science article** for University of Melbourne's *Pursuit* magazine: *Understanding how a cell becomes a person with maths* (link)

Community engagement

2019 **AMSI CHOOSEMATHS Day Q&A Panellist** - Raising mathematical enjoyment and aspirations of girls in secondary school years 9 & 10

2017–2018 **Open Day volunteer**, La Trobe University

Referees

Prof Jennifer Flegg, FAA
Director of ARC Centre of Excellence for MACSYS
University of Melbourne, VIC
jennifer.flegg@unimelb.edu.au

Ken A. Dill Endowed Prof Ramon Grima
Physical and Quantitative Biology
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Prof Peter Taylor, FAA
Chair of Operations Research
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